(the blue curve) has a more gradually changing slope. As a result, the two curves cross at two locations. There the planet can travel around Gargantua with the outward centrifugal force balancing the inward gravitational force.

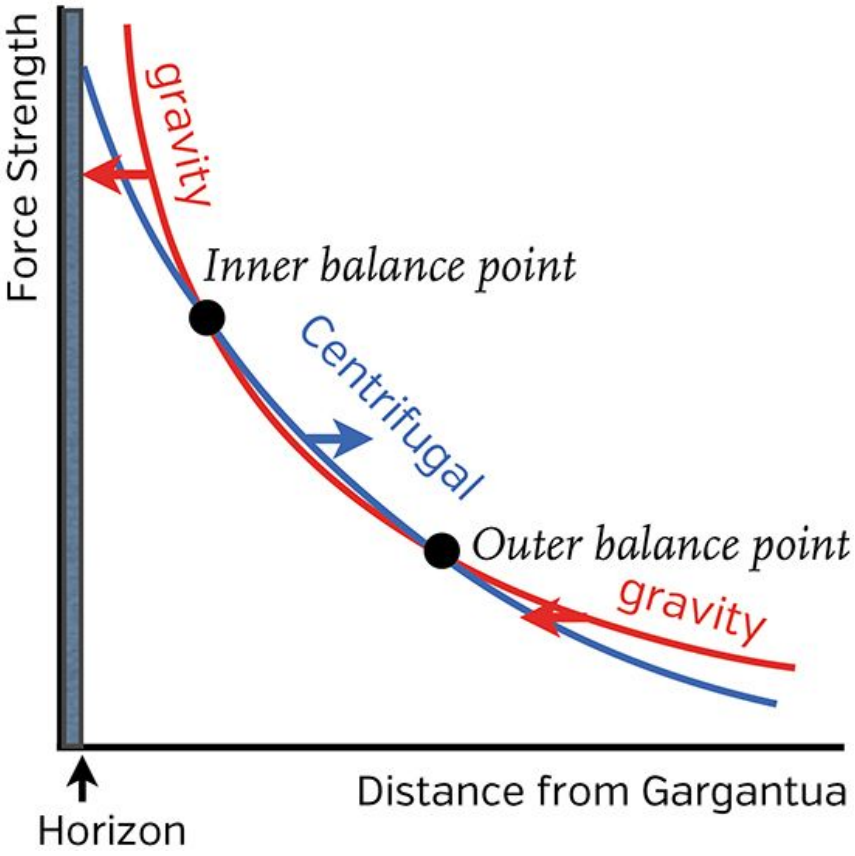


Fig. 17.2. The gravitational and centrifugal forces on Miller's planet.

At the inner balance point, the planet's orbit is unstable: If the planet gets pushed outward a tiny bit (for example, by the gravity of some passing comet), the centrifugal force wins the competition and pushes the planet further outward. If the planet is pushed inward, the gravitational force wins and the planet is pulled into Gargantua. This means Miller's planet can't live for long at the inner balance point.

The outer balance point, by contrast, is stable: If Miller's planet is there and gets pushed outward, gravity wins the competition and pulls